

Phase 2: Mathematical Model

Method Chosen: Newton's Second Law of Motion (Newton-Euler Formulation)

Justification: While Lagrange's equations are excellent for energy-based systems, Newton's Laws are chosen here because they explicitly expose the internal reaction forces (specifically, the cable tension). In industrial crane design, knowing the real-time tension is critical for cable sizing and safety factors. Furthermore, breaking the system down into individual free-body diagrams makes the physical coupling between the motors and the payload swing highly intuitive.

1. Coordinate System & Small Angle Assumption

- We define the Z-axis as positive downwards (direction of gravity).
- To make the step-by-step derivation manageable and realistic for control design, we apply the Small Angle Approximation for the payload swing. In industrial cranes, swing angles rarely exceed 10-15 degrees.
- Therefore: $\sin(\theta)$ is approximately equal to θ , and $\cos(\theta)$ is approximately equal to 1.

2. Kinematic Equations (Position, Velocity, Acceleration)

First, we define the absolute position of the payload mass (m_p) in 3D space. It depends on the bridge position (x), trolley position (y), cable length (L), and the small swing angles (θ_x and θ_y).

- Payload Position (X_p, Y_p, Z_p):
 - i. $X_p = x + L * \theta_x$
 - ii. $Y_p = y + L * \theta_y$
 - iii. $Z_p = L$

Next, we take the first derivative with respect to time to get the velocities. We must use the product rule since the cable length (L) can change while hoisting.

- Payload Velocity (V_{px}, V_{py}, V_{pz}):
 - i. $V_{px} = \dot{x} + (\dot{L} * \theta_x) + (L * \dot{\theta}_x)$
 - ii. $V_{py} = \dot{y} + (\dot{L} * \theta_y) + (L * \dot{\theta}_y)$
 - iii. $V_{pz} = \dot{L}$

Taking the second derivative gives the absolute accelerations.

- Payload Acceleration (A_{px} , A_{py} , A_{pz}):
 - i. $A_{px} = \ddot{x} + (\ddot{L} * \theta_x) + (2 * \dot{L} * \dot{\theta}_x) + (L * \ddot{\theta}_x)$
 - ii. $A_{py} = \ddot{y} + (\ddot{L} * \theta_y) + (2 * \dot{L} * \dot{\theta}_y) + (L * \ddot{\theta}_y)$
 - iii. $A_{pz} = \ddot{L}$

3. Applying Newton's Second Law (Sum of Forces = Mass * Acceleration)

We now analyze the Free Body Diagram for each mass component.

Step A: The Payload (m_p) in the Z-axis (Vertical)

Forces acting on the payload in the Z direction: Gravity pulling down ($+m_p * g$) and Cable Tension pulling up ($-T$).

Equation: $m_p * A_{pz} = m_p * g - T$

Substitute A_{pz} : $m_p * \ddot{L} = m_p * g - T$

Rearranging for Tension (T): $T = m_p * (g - \ddot{L})$

Step B: The Payload (m_p) in the X-axis and Y-axis (Swing)

Forces acting on the payload in the horizontal directions come entirely from the horizontal components of the cable tension. For small angles, the horizontal tension is $T * \theta$. Because tension pulls the payload back towards the vertical center, it is negative.

X-Axis Equation: $m_p * A_{px} = -T * \theta_x$

Y-Axis Equation: $m_p * A_{py} = -T * \theta_y$

Step 3C: Deriving the Swing Equations of Motion

Let's substitute our A_{px} and T equations into the X-Axis Payload equation.

- $m_p * [\ddot{x} + (\ddot{L} * \theta_x) + (2 * \dot{L} * \dot{\theta}_x) + (L * \ddot{\theta}_x)] = - [m_p * (g - \ddot{L})] * \theta_x$

Divide both sides by mass (m_p) to simplify:

- $\ddot{x} + (\ddot{L} * \theta_x) + (2 * \dot{L} * \dot{\theta}_x) + (L * \ddot{\theta}_x) = -g * \theta_x + (\ddot{L} * \theta_x)$

Notice that the term $(\ddot{L} * \theta_x)$ exists on both sides and cancels out. Rearranging the remaining terms gives us our final payload swing equation for the X-axis. The Y-axis follows the exact same symmetrical logic.

4. Final Equations of Motion (EOM)

EOM 1: Payload Swing in X-Z Plane

- $L * \ddot{\theta}_x + 2 * \dot{L} * \dot{\theta}_x + g * \theta_x = -\ddot{x}$

(Physical Meaning: This reveals that the bridge acceleration, $\ddot{x}$, acts as a driving force that causes the pendulum to swing. The term $2 * \dot{L} * \dot{\theta}_x$ is the Coriolis acceleration that occurs when the payload is hoisted while swinging.)

EOM 2: Payload Swing in Y-Z Plane

- $L * \ddot{\theta}_y + 2 * \dot{L} * \dot{\theta}_y + g * \theta_y = -\ddot{y}$

(Physical Meaning: Mirrors the X-axis; the trolley acceleration acts as the external disturbance causing the payload to oscillate in the Y direction.)

EOM 3: Bridge Dynamics (Mass = M_b)

Forces: Motor Force (F_x), Friction opposing motion ($-c_x * \dot{x}$), and the horizontal pull of the cable tension from the swinging payload ($+T * \theta_x$).

- $M_b * \ddot{x} = F_x - c_x * \dot{x} + m_p * (g - \ddot{L}) * \theta_x$

(Physical Meaning: To move the bridge, the motor must overcome the bridge inertia, track friction, and the reactive tugging force of the swinging payload.)

EOM 4: Trolley Dynamics (Mass = m_t)

Forces: Motor Force (F_y), Friction ($-c_y * \dot{y}$), and cable tension horizontal pull ($+T * \theta_y$).

- $m_t * \ddot{y} = F_y - c_y * \dot{y} + m_p * (g - \ddot{L}) * \theta_y$

(Physical Meaning: The trolley motor must overcome trolley inertia, friction, and the Y-axis pendulum reaction force.)

EOM 5: Hoist Dynamics

If we assume the hoist motor directly controls the vertical force (F_z) acting on the cable:

- $m_p * \ddot{L} = m_p * g - F_z$

(Physical Meaning: If the hoist motor force equals the weight of the payload, acceleration is zero and it hoists at a constant velocity. If it drops below payload weight, the load accelerates downward.)

5. Units Consistency Check

Let us verify EOM 1: $L * \ddot{\theta}_x + 2 * \dot{L} * \dot{\theta}_x + g * \theta_x = -\ddot{x}$

- $L * \ddot{\theta}_x$: (meters) * (1 / seconds²) = m/s²
- $2 * \dot{L} * \dot{\theta}_x$: (meters / second) * (1 / second) = m/s²
- $g * \theta_x$: (m/s²) * (dimensionless radians) = m/s²
- $-\ddot{x}$: m/s²

All terms share the unit of linear acceleration (m/s²), confirming dimensional consistency across the derived dynamic model.